

COS 135 Individual Assignment 5

Due: Monday 03/02/26 End of the day

This assignment has 3 C programs to be developed.

What to submit:

- .c source codes for each task.
- Your programs must produce similar outputs as given if the same inputs are provided.

[10 pts] Comments are required in the following locations (in each C source code):

- At the top of the source code comment your name and a short program description.
- Comment the purpose of each variable.
- Comment major sections of code such as input, processing, and output.

Program Design: Your program is a professional document and must be neat and easy to read. All programs should follow these specifications.

- Comments should be aligned and entered in a consistent fashion
- Blank lines should be added to aid readability
- Code within blocks should be indented
- Comments should not contain spelling mistakes
- Variables names should be meaningful

Write C programs for following tasks and submit your source codes (**you must submit .c files without compilation warnings or errors**).

Use the -Wall switch in gcc to enable all warnings:

\$ sudo gcc hello.c -o hello -Wall

Sample Program inputs are highlighted in **yellow**.

(a) [20 pts]

Write a C program to continuously check whether an alphabet (lowercase or uppercase) entered by the user is a vowel or a consonant. Enter # to exit the program.

First your program has to verify the character entered by the user is part of the English alphabet (can be lowercase or uppercase). If yes, check it for a vowel or a constant.

<u>Sample output # 1:</u>	<u>Sample output # 2:</u>	<u>Sample output # 3:</u>
Enter a letter: a a is a vowel	Enter a letter: N N is a consonant.	Enter a letter: \$ '\$' is not a valid character

Example program execution:

```
nimesha@cos135:/media/cos135/a5_codes$ sudo gcc hw5a_1.c -o hw5a
nimesha@cos135:/media/cos135/a5_codes$ ./hw5a
Enter a letter: Q
'Q' is a consonant
Enter a letter: E
'E' is a vowel
Enter a letter: i
'i' is a vowel
Enter a letter: a
'a' is a vowel
Enter a letter: @
'@' is not a valid character
Enter a letter: 2
'2' is not a valid character
Enter a letter: A
'A' is a vowel
Enter a letter: g
'g' is a consonant
Enter a letter: #
'#' is not a valid character
Good bye!
nimesha@cos135:/media/cos135/a5_codes$
```

(b) [30 pts]

Write a C program that can convert a decimal number to its binary equivalent and a binary number to its decimal equivalent.

Your program should test for inputs between 0 and 1000.

Note: you are not permitted to use arrays or other concepts not yet covered in this class.

<u>Sample output # 1:</u> Enter a binary or decimal number: b1101 Decimal value of the binary number 1101 is 13	<u>Sample output # 3:</u> Enter a binary or decimal number: c123 Not a valid input.
<u>Sample output # 2:</u> Enter a binary or decimal number: d245 Binary value of the decimal number 245 is 11110101	<u>Sample output # 4:</u> Enter a binary or decimal number: b104 Not a valid input.

Following are two sample algorithms (pseudo codes) to convert decimal numbers to binary and binary to decimal. **Note that error handling is not included.**

<u>Algorithm for Decimal to Binary conversion:</u> <pre>begin: read (decimal); binary ← 0; place ← 1; while (decimal > 0) do begin binary ← binary + ((decimal % 2) * place); place ← place * 10; decimal ← decimal / 2; end write('Binary number is: ' binary) end</pre>	<u>Algorithm Binary to Decimal conversion:</u> <pre>begin: read (binary); decimal ← 0; place ← 1; loop_pass ← 1; while (binary > 0) do begin if (loop_pass == 1) then place ← place * 1; else place ← place * 2; end if decimal ← decimal + (binary % 10) * place; binary ← binary / 10; loop_pass ← loop_pass + 1 end write('Decimal number is: ' decimal) end</pre>
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Example program execution:

```
nimesha@cos135:/media/cos135/a5_codes$ sudo gcc hw5b.c -o hw5b
nimesha@cos135:/media/cos135/a5_codes$ ./hw5b
Enter a binary or decimal number:d843
Binary value of the decimal number 843 is 1101001011
nimesha@cos135:/media/cos135/a5_codes$ ./hw5b
Enter a binary or decimal number:d 499
Binary value of the decimal number 499 is 111110011
nimesha@cos135:/media/cos135/a5_codes$ ./hw5b
Enter a binary or decimal number:b10010011
Decimal value of the binary number 10010011 is 147
nimesha@cos135:/media/cos135/a5_codes$ ./hw5b
Enter a binary or decimal number:b 1101111
Decimal value of the binary number 1101111 is 111
nimesha@cos135:/media/cos135/a5_codes$ ./hw5b
Enter a binary or decimal number:c234
Not a valid input.
nimesha@cos135:/media/cos135/a5_codes$ ./hw5b
Enter a binary or decimal number:b1234
Not a valid binary number.
nimesha@cos135:/media/cos135/a5_codes$
```

(c) **[40 pts]** This program will help a student to stay within a monthly budget. First, ask the user for the monthly expense goal. Next, ask the user for the amount spend on rent, utilities, transportation, and food. **Assume an additional fixed expense of a car payment of \$581.99 that should be set up as a variable.**

As shown below, report in a neat chart with each item, its amount and percentage (% of total expenses). Under the chart, report the goal, the total expenses, and the amount leftover. Precision **should be set to two decimal places**. The chart **should use field width** (not multiple blanks) to neatly align columns. The chart should **include your name** (replace 'COS135' with your name) in the chart heading and should use blank lines as shown to separate content.

Finally,

- if the leftover amount is greater than or equal to $1/5^{\text{th}}$ of the expense goal (i.e., at least \$500 from \$2500) display "Good job. You are saving enough!"
- if the leftover amount is less than $1/5^{\text{th}}$ of the expense goal (i.e., less than \$500 from \$2500) display "Try to save more!"

Test your program on several data sets.

Use the following datasets to verify the final submission.

```
nimesha@cos135:/media/cos135/a5_codes$ sudo gcc hw5c.c -o hw5c
[sudo] password for nimesha:
nimesha@cos135:/media/cos135/a5_codes$ ./hw5c
Enter monthly expense goal (USD): 3600
Enter rent (USD): 500
Enter utilities (USD): 400
Enter transportation (USD): 200
Enter food costs (USD): 700

Budget Chart for COS135:
Expense          Amount $          Percent %
Car payment      581.99           24.43
Rent             500.00           20.99
Utilities        400.00           16.79
Transportation   200.00           8.40
Food             700.00           29.39

Expense Goal      3600.00
Total expenses    2381.99
Leftover amount   1218.01
Good job. You are saving enough!
nimesha@cos135:/media/cos135/a5_codes$
nimesha@cos135:/media/cos135/a5_codes$
```

```
nimesha@cos135:/media/cos135/a5_codes$ ./hw5c
Enter monthly expense goal (USD): 2500
Enter rent (USD): 700
Enter utilities (USD): 450.50
Enter transportation (USD): 175.50
Enter food costs (USD): 400.00

Budget Chart for COS135:
Expense          Amount $          Percent %
Car payment      581.99           25.22
Rent             700.00           30.33
Utilities        450.50           19.52
Transportation   175.50           7.60
Food             400.00           17.33

Expense Goal      2500.00
Total expenses    2307.99
Leftover amount   192.01
Try to save more!
nimesha@cos135:/media/cos135/a5_codes$
nimesha@cos135:/media/cos135/a5_codes$
nimesha@cos135:/media/cos135/a5_codes$
```